CS683 Programming Assignment 2

Stride-pattern prefetching and instruction-conscious L2 replacement

Setup

Every reported run is a single-core ChampSim simulation with the hashed perceptron predictor, built with GCC 7 (g++-7 7.5.0). Warmup is 25 million instructions and measurement is the following 25 million. The three traces are Trace 1, Trace 2, and Trace 3.

IPC is retired instructions divided by cycles in the region of interest. Speedup divides that IPC by the IPC of the same trace with no L1D prefetcher and LRU replacement. Load MPKI is the simulator's demand-load MPKI. Prefetch accuracy is the printed L1D ACCURACY: useful prefetches divided by prefetches sent to the next level. Coverage is the reduction in L1D demand-load misses against the no-prefetch LRU run. IPC, speedup, and MPKI are summarized by a geometric mean. Accuracy and coverage are summarized by an arithmetic mean.

Task 1. Stride pattern prefetcher

The stride pattern prefetcher trains at the L1D on demand loads. A 16-set, 4-way table, indexed by a hash of the full instruction pointer, tracks each IP. An entry holds the last cache-line address, the last stride, a saturating uniform-stride confidence, and a signature of the four most recent strides. Each stride is stored as a 7-bit signed value from -63 to +63. A separate 256-entry direct-mapped table maps the pair (IP, signature) to the next stride and a 2-bit confidence.

Uniform strides are preferred. Confidence rises when the observed stride repeats and falls back to 1 when it changes. At or above the uniform threshold the prefetcher repeats that stride. Otherwise, once four strides are in the signature, it looks up the pattern table and follows the stored next stride. Further degrees update a temporary signature, so a repeating sequence such as (+1, +2, +2, +2, +3) can be walked without training on the predictions themselves. Any prefetch that would leave the current 4 KB page is dropped, and a stride outside -63..+63 clears the history for that IP.

The reported configuration is the best of the 12-point sweep: uniform confidence 2, pattern confidence 1, and accuracy boundaries 1%, 2%, and 4%. Those parameters are selected at run time. The source defaults remain uniform confidence 4, pattern confidence 2, and boundaries 25/50/75.

Storage

The table below counts compact state, not the padded C++ objects in the simulator. A full 64-bit IP is charged because the code compares the whole pointer. The set index is a hash, so it is not stored again.

Structure	Entry	Bits / entry	Entries	Bits
IP tracker	valid, full IP, last line, stride, confidence, signature, history, LRU	166	64	10,624
Pattern table	valid, IP, signature, next stride, confidence	102	256	26,112
Controller	interval, degree, three thresholds	42	1	42

Compact state is 36,778 bits, about 4.49 KiB, when the cache already exposes issued and useful counters. The supplied IP-stride table is 24 entries x 130 bits = 3,120 bits.

The degree register holds the values 1, 2, 4, and 8. The 12-bit demand counter walks 0..4095 and then resets. If the cache does not already expose issued and useful counts, those two 64-bit counters add 128 bits. The uniform-prediction and pattern-prediction counters are diagnostics and are not part of the predictor.

Thresholds and degree

Degree starts at 2. Every 4096 demand loads, once at least 128 prefetches have been issued, degree is chosen from useful prefetches divided by all issued prefetches. That ratio counts requests that hit a line already present, so it is harsher than the printed accuracy and reacts to wasted issue bandwidth. Below 1% the degree is 1, below 2% it is 2, below 4% it is 4, and otherwise it is 8.

The steps are powers of two so each accuracy band looks one step farther along the same stride or signature. Degree 1 is the safe next line when the prefetcher is mostly missing. Degree 8 is the largest lookahead tested; it is used only after the issued stream has stayed useful. On the selected runs the final degree is 8 on Trace 1, 4 on

Trace 2, and 4 on Trace 3.

Uniform confidence 2 starts the constant-stride path after two repeats, which is early enough for short runs of the same stride and still gives the pattern table the accesses that break the constant. Pattern confidence 1 accepts a next stride after one confirmation. Both choices were tested against the stricter defaults. The strict defaults finish near degree 1 and gain only about two tenths of a percent. The looser gates issue more prefetches, give up some printed accuracy, and win IPC and coverage on every trace. The ranking is the reason for the reported thresholds; it is specific to these three traces.

Configuration	Speedup	L1D MPKI	Accuracy	Coverage
u=2, p=1, a=1/2/4	+0.60%	7.447	45.83%	5.63%
u=2, p=2, a=1/2/4	+0.56%	7.471	45.03%	5.34%
u=2, p=1, a=3/6/12	+0.48%	7.569	52.11%	4.10%
u=2, p=2, a=3/6/12	+0.40%	7.596	50.60%	3.76%
u=2, p=1, a=25/50/75	+0.40%	7.644	51.90%	3.16%
u=2, p=2, a=25/50/75	+0.37%	7.663	50.93%	2.92%
u=4, p=1, a=1/2/4	+0.34%	7.626	54.84%	3.39%
u=4, p=1, a=3/6/12	+0.29%	7.693	60.35%	2.55%
u=4, p=2, a=1/2/4	+0.27%	7.667	54.60%	2.86%
u=4, p=1, a=25/50/75	+0.25%	7.728	59.21%	2.10%
u=4, p=2, a=3/6/12	+0.24%	7.723	60.12%	2.16%
u=4, p=2, a=25/50/75	+0.21%	7.758	58.62%	1.73%

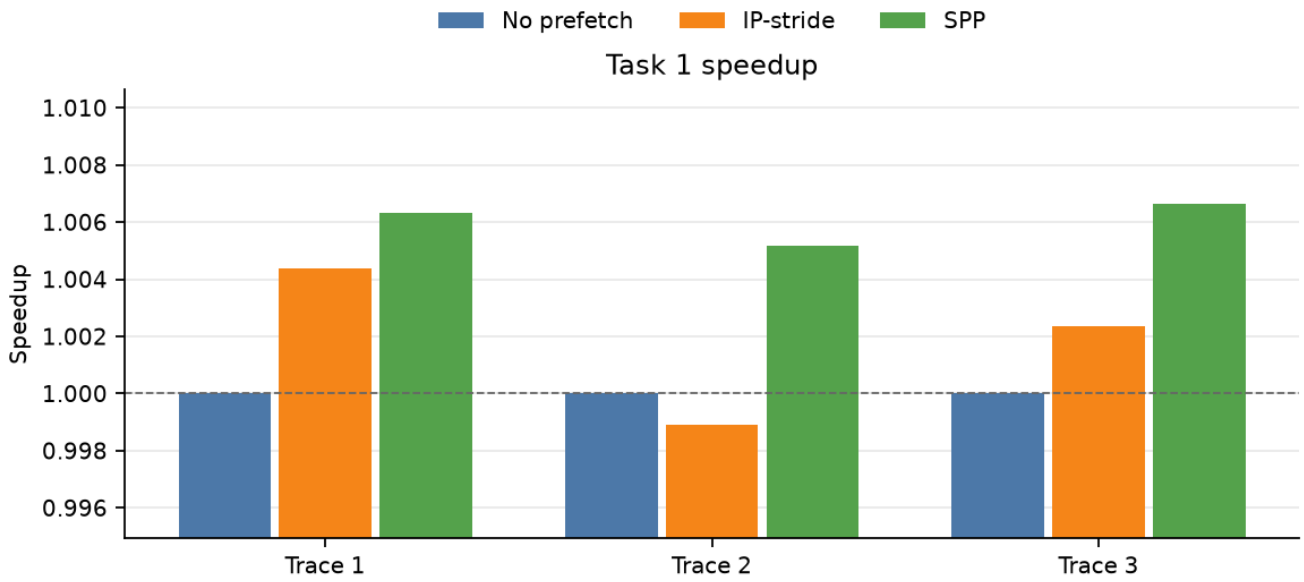
Geometric mean of IPC speedup and L1D load MPKI. Arithmetic mean of accuracy and coverage. Sorted by speedup. The first row is the configuration used below and in Task 3.

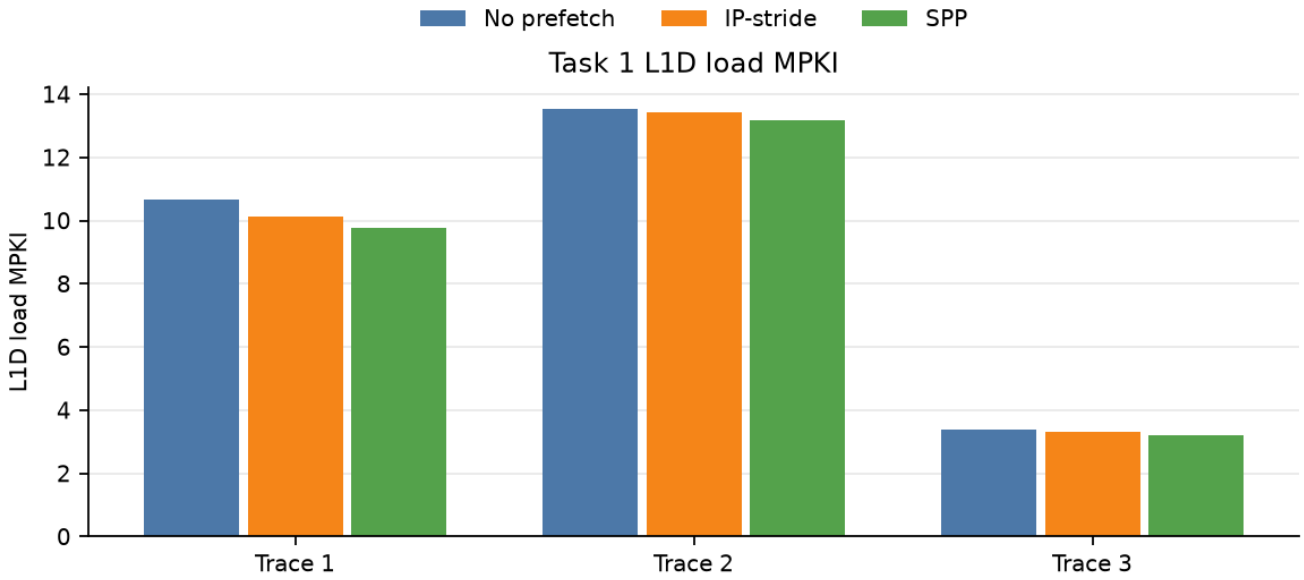
Task 1 results

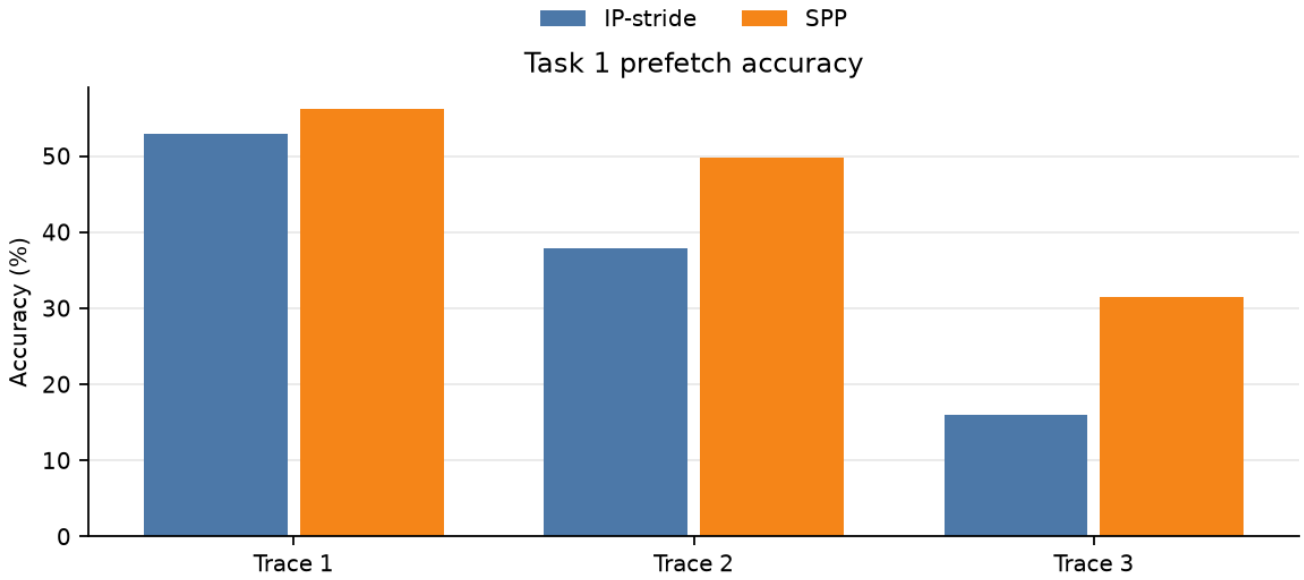
Selected SPP gains 0.604% geometric-mean IPC over no prefetching, compared with 0.188% for the supplied IP-stride prefetcher. It also lowers L1D load MPKI on every trace. Printed accuracy stays above IP-stride even in this more aggressive setting.

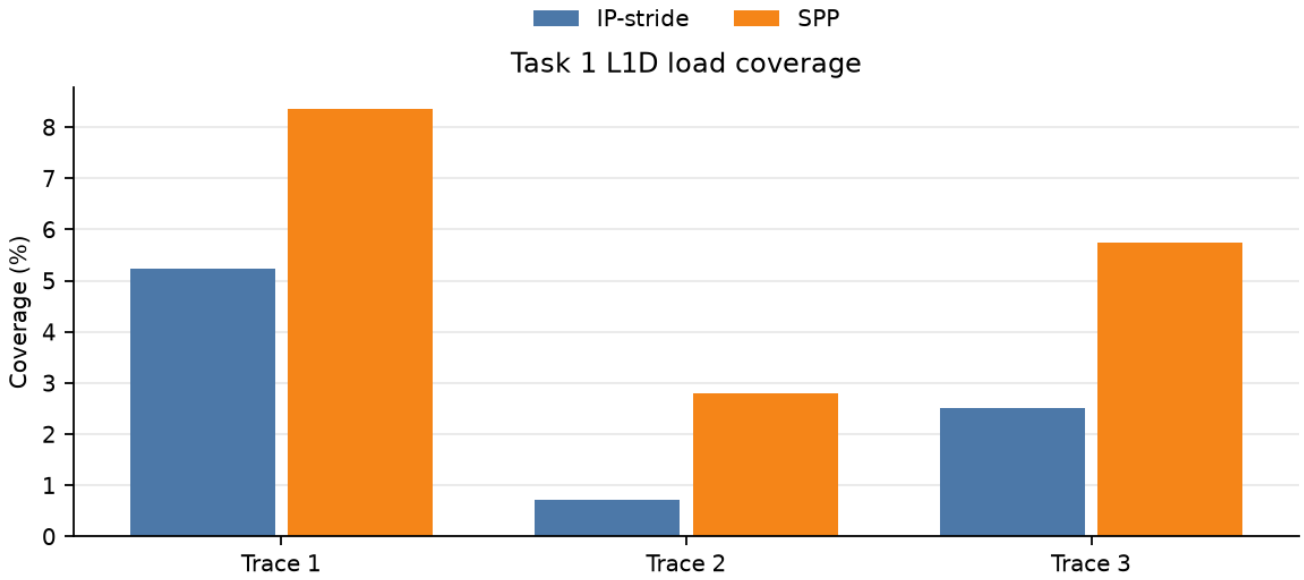
Trace	Config	IPC	Speedup	L1D MPKI	Accuracy	Coverage
Trace 1	No prefetch	0.2318	+0.00%	10.678	-	-
Trace 1	IP-stride	0.2328	+0.44%	10.118	52.94%	5.25%
Trace 1	SPP	0.2333	+0.63%	9.786	56.20%	8.36%
Trace 2	No prefetch	0.2515	+0.00%	13.537	-	-
Trace 2	IP-stride	0.2513	-0.11%	13.438	37.88%	0.73%
Trace 2	SPP	0.2528	+0.52%	13.158	49.78%	2.80%
Trace 3	No prefetch	0.3818	+0.00%	3.404	-	-
Trace 3	IP-stride	0.3827	+0.23%	3.318	16.06%	2.52%
Trace 3	SPP	0.3844	+0.66%	3.208	31.51%	5.75%
Mean	**No prefetch**	**0.2813**	**+0.00%**	**7.894**	**-**	**-**
Mean	**IP-stride**	**0.2819**	**+0.19%**	**7.670**	**35.63%**	**2.83%**
Mean	**SPP**	**0.2830**	**+0.60%**	**7.447**	**45.83%**	**5.63%**

Mean IPC, speedup, and MPKI are geometric. Mean accuracy and coverage are arithmetic. Baseline accuracy is undefined because it issues no prefetches.


No prefetch
IP-stride
SPP
Task 1 speedup
Speedup
1.010
1.008
1.006
1.004
1.002
1.000
0.998
0.996
Trace 1
Trace 2
Trace 3


No prefetch
IP-stride
SPP
Task 1 L1D load MPKI
L1D load MPKI
14
12
10
8
6
4
2
0
Trace 1
Trace 2
Trace 3





Task 2A. Instruction-conscious RRIP

I-RRIP is SRRIP with a sticky instruction bit on each L2 line. A fill is marked as an instruction only when the incoming packet is an instruction load or instruction prefetch. Hits keep the resident bit, including a line that is later touched as data. Translation requests do not mark a line as an instruction. RRPV still uses two bits, with 3 meaning ready for replacement.

Insertion is unchanged from SRRIP: instruction and data fills both enter at RRPV 2. The sweep's best six configurations all insert at 2. Inserting instructions at 0 or 1 keeps them too long before they have shown reuse, and the best insertion-1 configuration is well behind the winner. Promotion is where instructions are preferred. An instruction hit sets RRPV to 0. A data hit sets RRPV to 1, so reused data is not immediately replaceable but remains behind a re-referenced instruction. Giving data hits RRPV 0 as well was slightly worse at the best eviction settings.

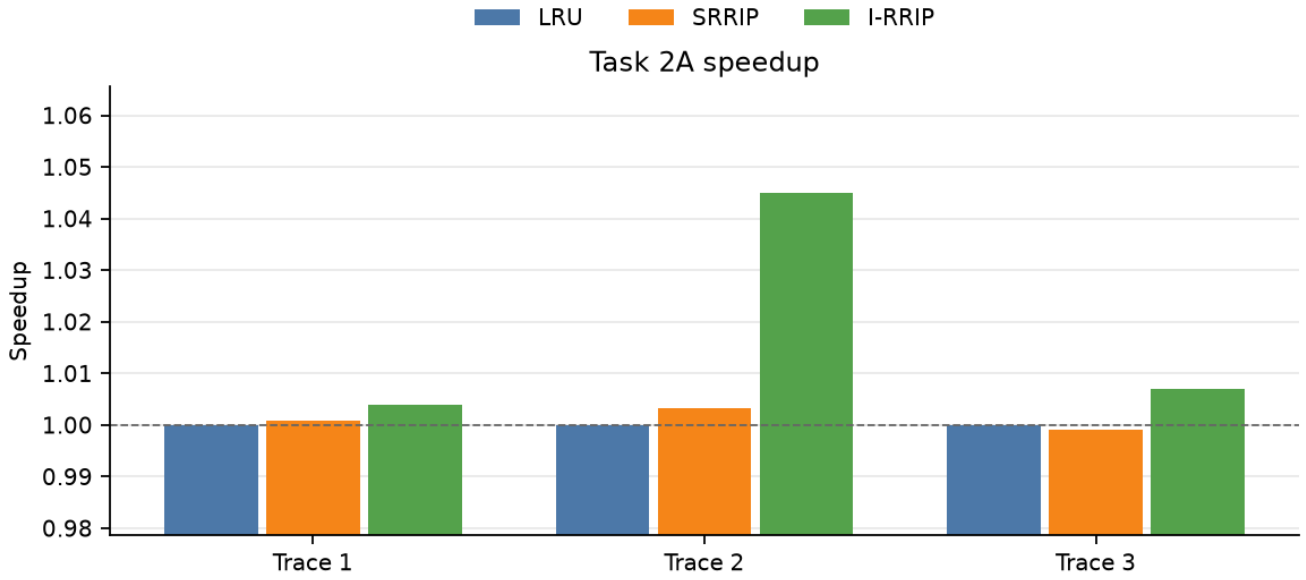
Eviction still takes an invalid way first, then ages every line until some RRPV reaches 3. With bias 2, if any data line has RRPV at least 1, the victim is the data line with the highest RRPV; ties keep the earlier way. A data line at RRPV 0 is left alone, so a just-filled or just-hit data line can survive one aging step. Bias 3, which always prefers data when any data line exists, tied this result exactly. Bias 2 is kept because it is the less aggressive rule and matches the best IPC. Turning awareness off reproduces SRRIP and loses almost all of the gain.

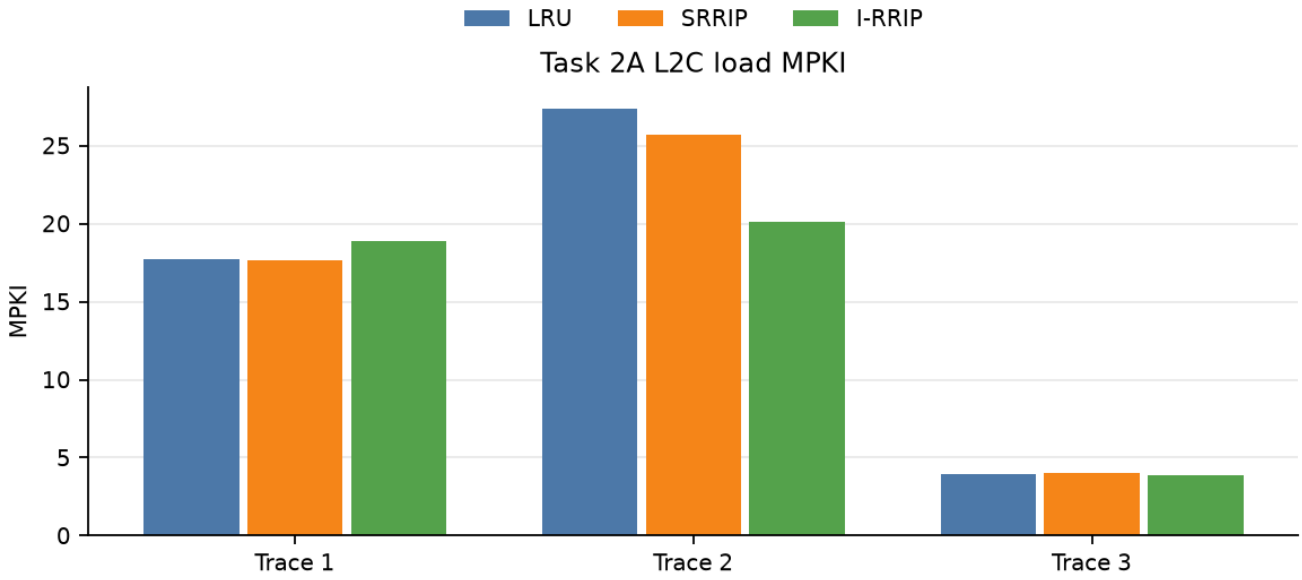
The IPC gain is 1.844% over LRU. Most of it is on Trace 2, where L2 instruction MPKI falls from 18.809 to 7.719. Data MPKI rises on every trace, which is the cost of holding instructions. Overall L2 load MPKI still falls on Trace 2

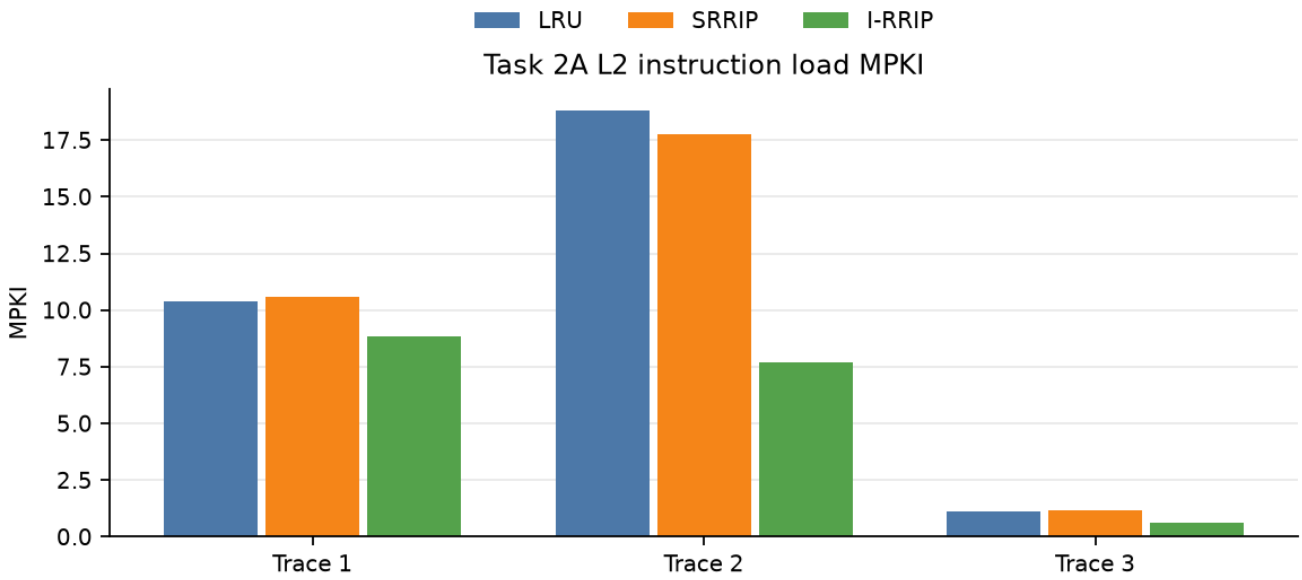
and Trace 3.

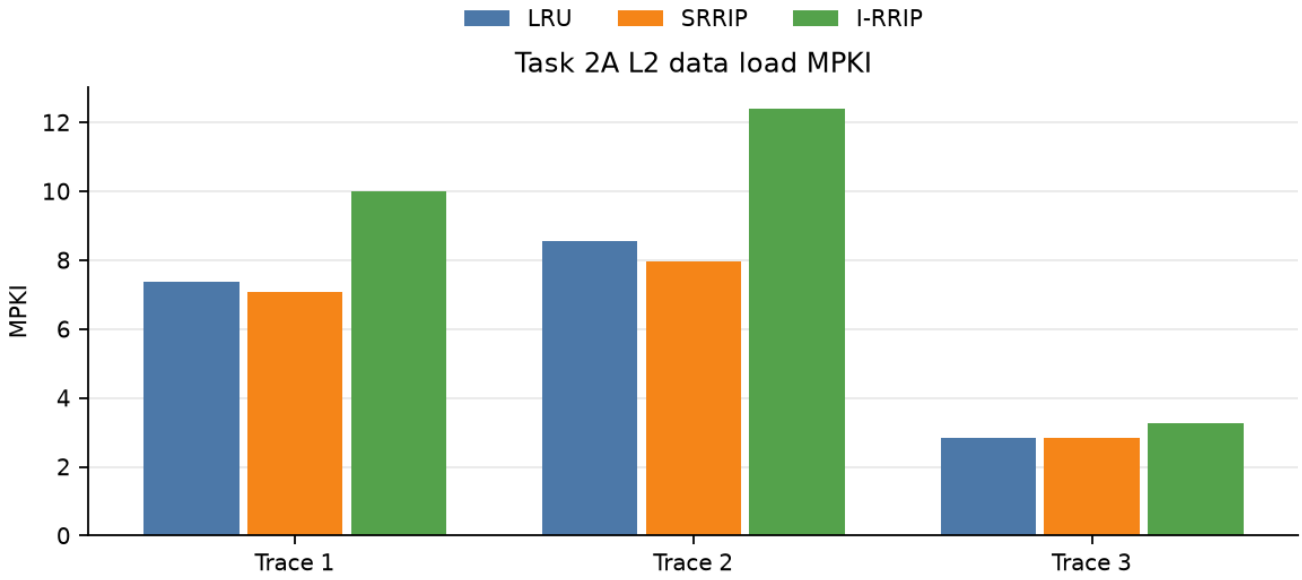
Trace	Policy	IPC	Speedup	L2C MPKI	L2 inst. MPKI	L2 data MPKI
Trace 1	LRU	0.2318	+0.00%	17.724	10.363	7.361
Trace 1	SRRIP	0.2320	+0.08%	17.657	10.590	7.066
Trace 1	I-RRIP	0.2327	+0.39%	18.883	8.866	10.017
Trace 2	LRU	0.2515	+0.00%	27.373	18.809	8.565
Trace 2	SRRIP	0.2524	+0.33%	25.730	17.766	7.965
Trace 2	I-RRIP	0.2629	+4.50%	20.123	7.719	12.404
Trace 3	LRU	0.3818	+0.00%	3.954	1.119	2.836
Trace 3	SRRIP	0.3815	-0.08%	4.034	1.189	2.846
Trace 3	I-RRIP	0.3845	+0.69%	3.902	0.624	3.278
Mean	**LRU**	**0.2813**	**+0.00%**	**12.426**	**6.019**	**5.633**
Mean	**SRRIP**	**0.2816**	**+0.11%**	**12.238**	**6.070**	**5.431**
Mean	**I-RRIP**	**0.2865**	**+1.84%**	**11.403**	**3.495**	**7.413**

Speedup is against no-prefetch LRU. Mean IPC, speedup, and MPKI are geometric. Instruction and data MPKI are L2 demand-load misses.




LRU
SRRIP
I-RRIP
Task 2A L2C load MPKI
MPKI
0
5
10
15
20
25
Trace 1
Trace 2
Trace 3


LRU
SRRIP
I-RRIP
Task 2A L2 instruction load MPKI
MPKI
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Trace 1
Trace 2
Trace 3



Task 2B. Dynamic I-RRIP

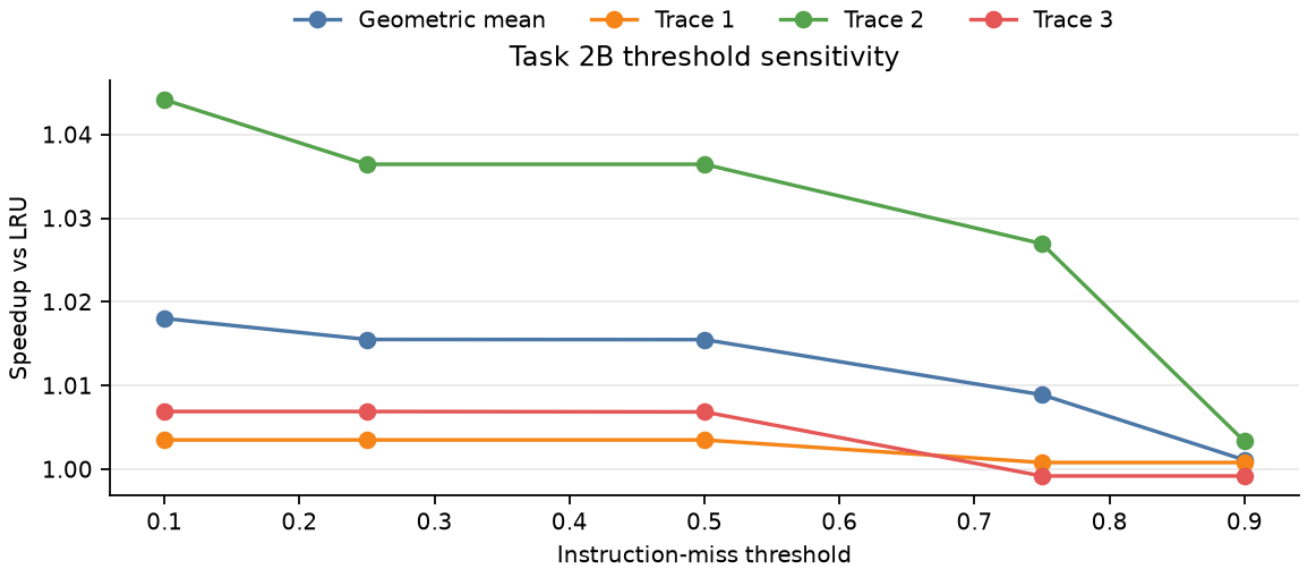
Dynamic I-RRIP uses the same insertion, promotion, and bias as the selected static policy, but only on follower sets, and only while a sampled miss share says instructions are the problem. Thirty-two of the 1024 sets are permanent SRRIP leaders. Set 33k, for k from 0 to 31, is the leader of each 32-set group, so the sample is spread through the cache instead of sitting in one region. Leaders are 3.125% of the sets: enough to observe instruction and data misses, and small enough that the other 992 sets can follow the decision.

Only demand loads train the counters, and the type comes from the incoming request rather than the sticky bit of the resident line. After every 4096 leader loads the controller computes instruction-load misses divided by all load misses. A share strictly above the threshold selects I-RRIP. Equality, and a window with no misses, stays on SRRIP. The four counters then reset. The decision applies to later accesses; it does not rewrite lines already in the cache. Learning runs through warmup and continues into the measured region.

The static threshold in the assignment hint is 0.25. The required sweep also measures 0.10, 0.50, 0.75, and 0.90. Lower thresholds protect instructions more often. On Trace 1 the three lowest thresholds make the same decision. On Trace 2, moving from 0.90 to 0.10 is the difference between staying near SRRIP and keeping the large instruction-MPKI reduction.

Threshold	Geomean	Trace 1	Trace 2	Trace 3
0.10	+1.80%	+0.35%	+4.42%	+0.69%
0.25	+1.55%	+0.35%	+3.64%	+0.69%
0.50	+1.55%	+0.35%	+3.64%	+0.68%
0.75	+0.89%	+0.08%	+2.69%	-0.08%
0.9	+0.11%	+0.08%	+0.33%	-0.08%

Speedup versus no-prefetch LRU. Five thresholds required by the sensitivity study.



Threshold 0.10 is the best tested value, at 1.802% geometric-mean IPC over LRU. Static I-RRIP is 1.844%. Dynamic I-RRIP is 0.041% behind that static policy. At 0.10 the follower sets use I-RRIP for the entire measured region on all three traces, so the remaining gap is the 32 SRRIP leaders and the warmup history. The source default stays 0.25. The reported dynamic runs set the threshold to 0.10. Two later controller variants, starting in I-RRIP and requiring two weak windows before leaving it, did not beat this result.

Trace	IPC	Speedup	L2C MPKI	L2 inst. MPKI	L2 data MPKI
Trace 1	0.2326	+0.35%	18.855	8.930	9.925
Trace 2	0.2626	+4.42%	20.114	8.001	12.113
Trace 3	0.3845	+0.69%	3.903	0.637	3.266

Dynamic I-RRIP at threshold 0.10, the best value in the sweep.

Policy	ROI mode	Speedup	L2C MPKI	L2 inst. MPKI	L2 data MPKI
LRU	-	+0.00%	12.426	6.019	5.633
I-RRIP	-	+1.84%	11.403	3.495	7.413
Dynamic 0.10	100% I-RRIP in ROI	+1.80%	11.397	3.571	7.322

Geometric means. Dynamic I-RRIP stays just behind static I-RRIP because 32 sets never leave SRRIP.

Task 3. SPP with dynamic I-RRIP

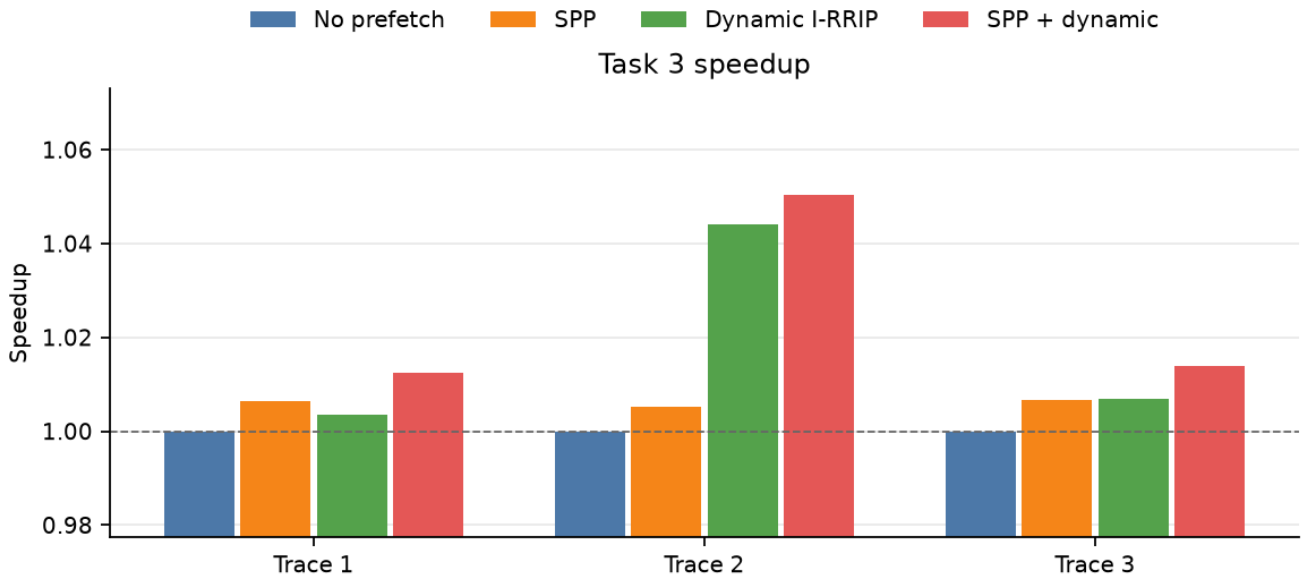
Task 3 keeps the selected SPP and the threshold-0.10 dynamic I-RRIP, with no joint retuning. One extra rule applies in follower sets when prefetch awareness is enabled. A data prefetch fill is inserted at RRPV 3, and a data prefetch hit does not change RRPV. A later demand hit still uses the current mode: RRPV 0 in SRRIP, or RRPV 1 for data and 0 for instructions in I-RRIP. Leader sets keep ordinary SRRIP, including for prefetches. Prefetches do not train the miss-share counters.

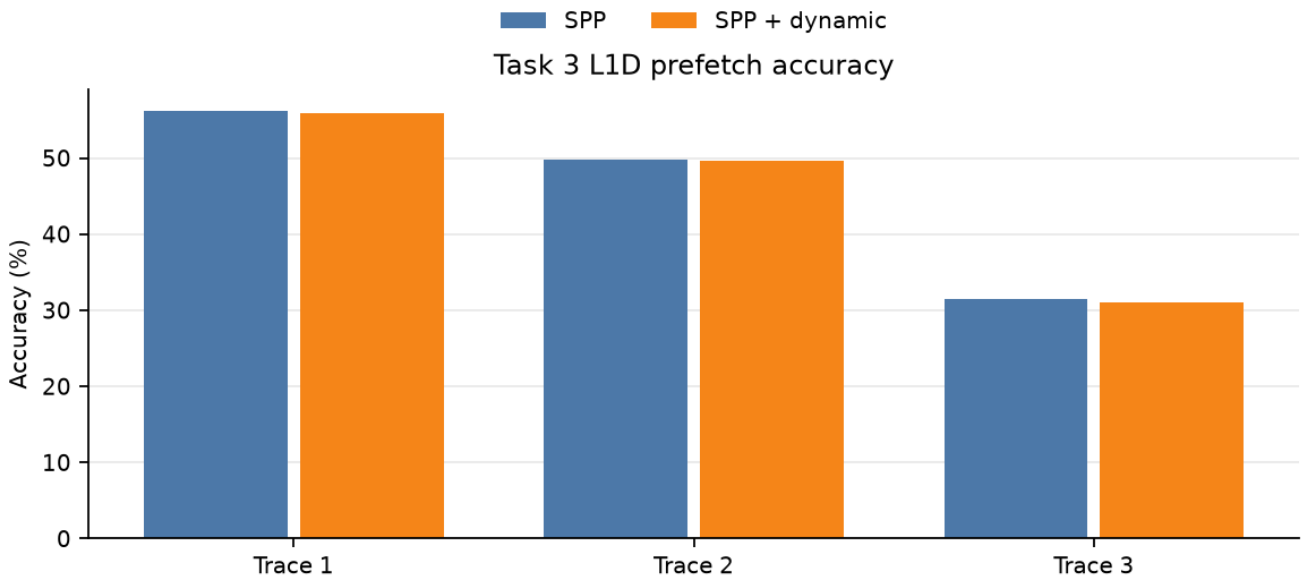
The reason is that a prefetched line has not been demanded yet. Inserting it near replacement lets a demand line or an instruction line win the victim choice, and ignoring prefetch hits stops a stream of speculative requests from refreshing a line that the program has not used. If the program does use it, the demand hit restores the normal priority. The risk is the opposite one: a useful prefetch can be evicted before the demand arrives. That is why the plain combination, with ordinary RRIP insertion of prefetches, is reported as well.

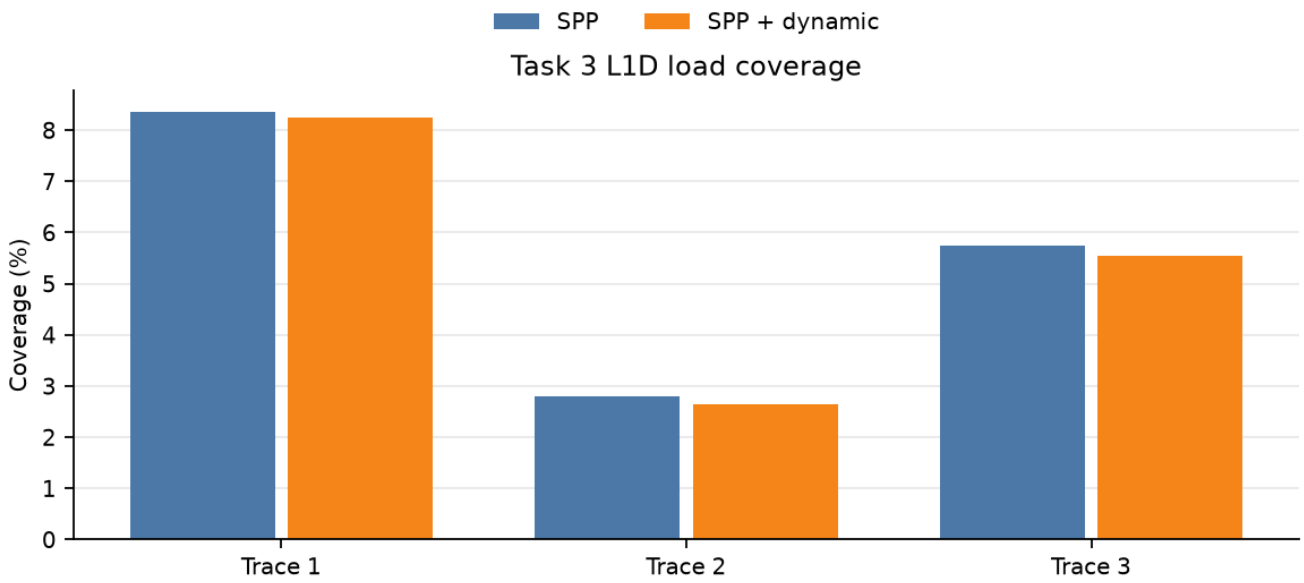
The prefetch-aware combination gains 2.548% geometric-mean IPC over no-prefetch LRU. Standalone SPP gains 0.604% and standalone dynamic I-RRIP gains 1.802%, so the combination beats both. The same policy without the prefetch rule gains 2.532%. Awareness adds a small further gain on Trace 1 and Trace 2. On Trace 3 the two combinations are effectively tied, with the plain combination ahead by a few thousandths of a percent. L2 instruction MPKI stays near the dynamic-I-RRIP level, while L1D MPKI and coverage stay near standalone SPP.

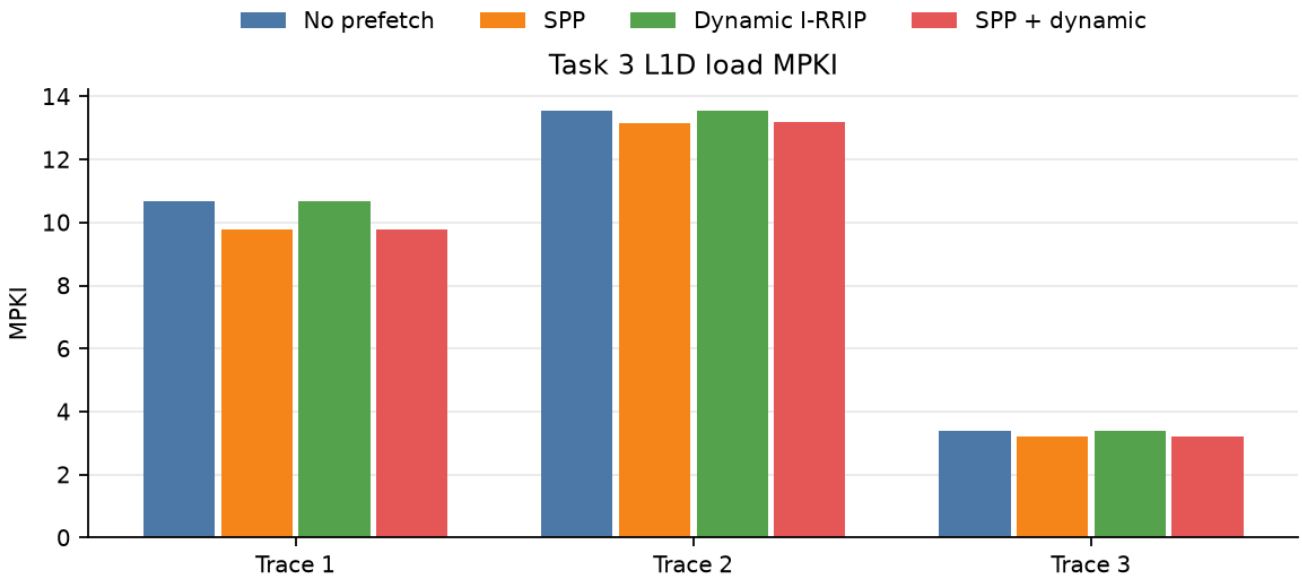
Trace	Config	IPC	Speedup	Acc.	Cov.	L1D	L2C	L2I	L2D
Trace 1	No prefetch	0.2318	+0.00%	-	-	10.678	17.724	10.363	7.361
Trace 1	SPP	0.2333	+0.63%	56.20%	8.36%	9.786	17.007	10.429	6.577
Trace 1	Dynamic I-RRIP	0.2326	+0.35%	-	-	10.679	18.855	8.930	9.925
Trace 1	SPP + dynamic	0.2347	+1.25%	55.93%	8.26%	9.797	17.918	8.916	9.002
Trace 2	No prefetch	0.2515	+0.00%	-	-	13.537	27.373	18.809	8.565
Trace 2	SPP	0.2528	+0.52%	49.78%	2.80%	13.158	27.368	19.165	8.203
Trace 2	Dynamic I-RRIP	0.2626	+4.42%	-	-	13.551	20.114	8.001	12.113
Trace 2	SPP + dynamic	0.2642	+5.04%	49.65%	2.65%	13.178	19.992	8.255	11.737
Trace 3	No prefetch	0.3818	+0.00%	-	-	3.404	3.954	1.119	2.836
Trace 3	SPP	0.3844	+0.66%	31.51%	5.75%	3.208	3.661	1.132	2.529
Trace 3	Dynamic I-RRIP	0.3845	+0.69%	-	-	3.404	3.903	0.637	3.266
Trace 3	SPP + dynamic	0.3872	+1.40%	31.04%	5.55%	3.215	3.594	0.639	2.955
Mean	**No prefetch**	**0.2813**	**+0.00%**	**-**	**-**	**7.894**	**12.426**	**6.019**	**5.633**
Mean	**SPP**	**0.2830**	**+0.60%**	**45.83%**	**5.63%**	**7.447**	**11.944**	**6.094**	**5.148**
Mean	**Dynamic I-RRIP**	**0.2864**	**+1.80%**	**-**	**-**	**7.898**	**11.397**	**3.571**	**7.322**
Mean	**SPP + dynamic**	**0.2885**	**+2.55%**	**45.54%**	**5.48%**	**7.459**	**10.878**	**3.610**	**6.784**

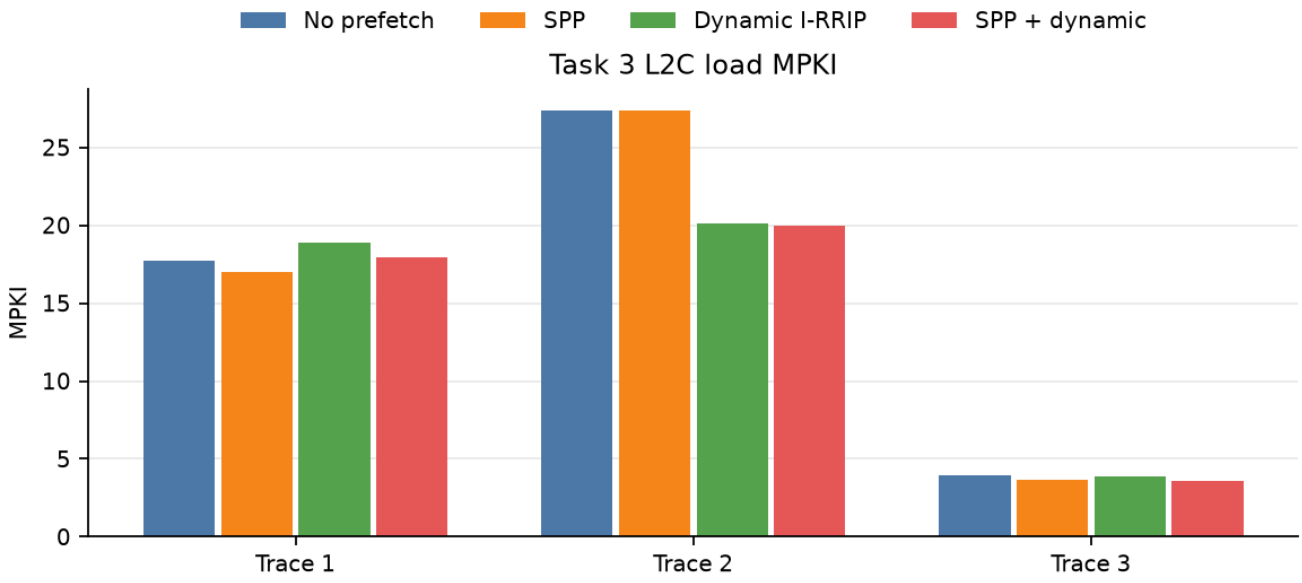
Accuracy and coverage refer to the L1D prefetcher. Dynamic I-RRIP alone issues no L1D prefetches. MPKI columns are demand-load MPKI. Means follow the same geometric and arithmetic split as Task 1.

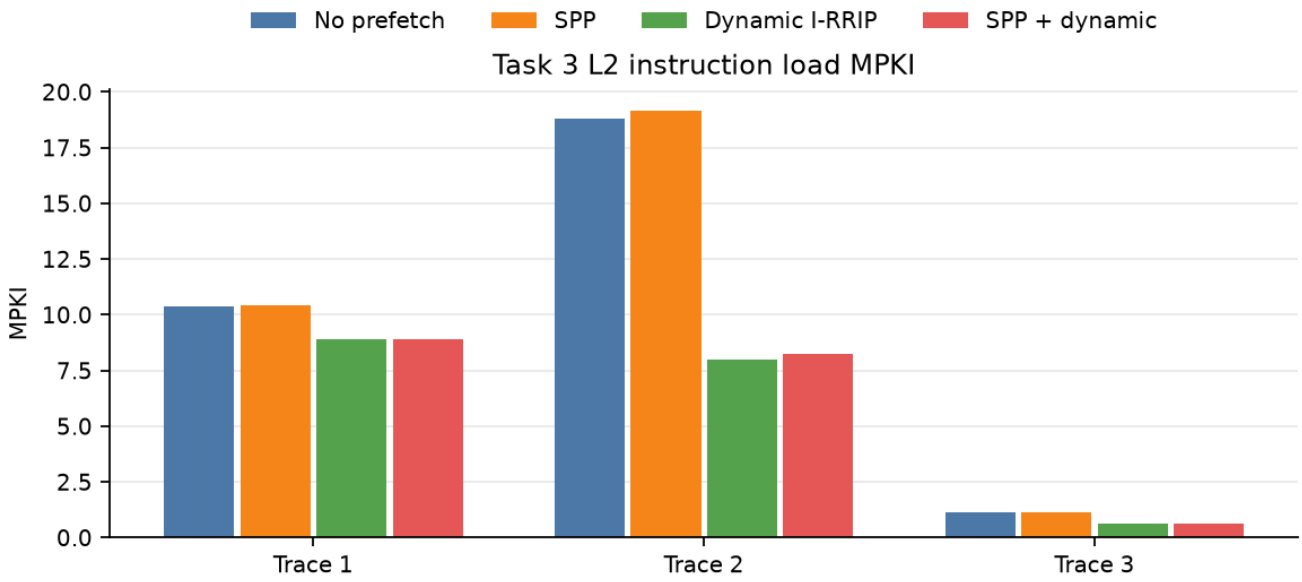


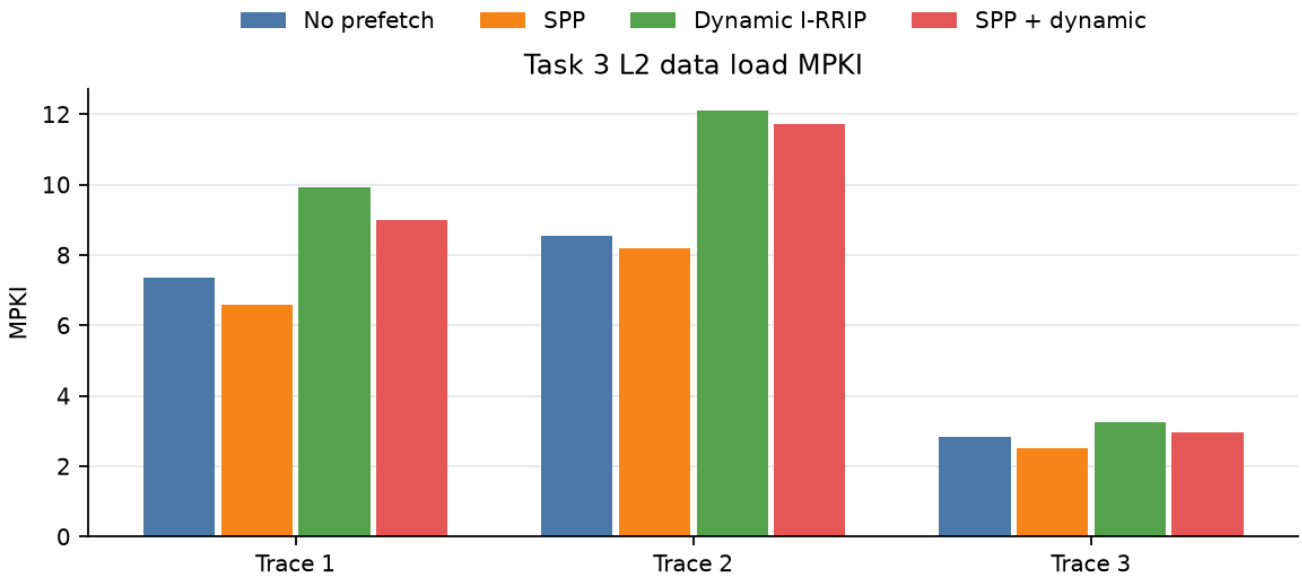

SPP
SPP + dynamic
Task 3 L1D prefetch accuracy
Accuracy (%)
0
10
20
30
40
50
Trace 1
Trace 2
Trace 3


SPP
SPP + dynamic
Task 3 L1D load coverage
Coverage (%)
0
1
2
3
4
5
6
7
8
Trace 1
Trace 2
Trace 3


No prefetch
SPP
Dynamic I-RRIP
SPP + dynamic
Task 3 L1D load MPKI
MPKI
0
2
4
6
8
10
12
14
Trace 1
Trace 2
Trace 3


No prefetch
SPP
Dynamic I-RRIP
SPP + dynamic
Task 3 L2C load MPKI
MPKI
0
5
10
15
20
25
Trace 1
Trace 2
Trace 3


No prefetch
SPP
Dynamic I-RRIP
SPP + dynamic
Task 3 L2 instruction load MPKI
MPKI
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Trace 1
Trace 2
Trace 3


No prefetch
SPP
Dynamic I-RRIP
SPP + dynamic
Task 3 L2 data load MPKI
MPKI
12
10
8
6
4
2
0
Trace 1
Trace 2
Trace 3